To create a CI/CD pipeline and deploy a **PHP Laravel application on Laravel Cloud** (also known as Laravel Forge or Laravel Vapor), you first need to choose whether you are using:

Option 1: Laravel Forge (for VPS/Cloud Server deployments)

Forge connects to DigitalOcean, AWS EC2, Linode, etc., and helps automate provisioning and deployment.

Option 2: Laravel Vapor (for serverless deployment on AWS Lambda)

Vapor is Laravel's own platform for deploying apps using AWS Lambda, S3, RDS, and more (serverless model).

I'll guide you through both options:

Laravel Forge Deployment + Pipeline (VPS Model)

1. Set up your Forge account and server

- Go to https://forge.laravel.com
- Connect your cloud provider (DigitalOcean, AWS, Linode, etc.)
- Create a server and provision it (Forge handles Nginx, PHP, MySQL, SSL)

2. Connect your Git repository

- On Forge, go to your server → **Sites** → **Create a site**
- Choose GitHub/GitLab/Bitbucket
- Select the repository of your Laravel app

3. Set environment variables

- Add .env variables in the Forge UI under **Environment**

4. Enable deployment scripts

Add a deployment script (Forge runs it every time you deploy):

```
cd /home/forge/yourdomain.com
git pull origin main
composer install --no-interaction --prefer-dist --optimize-autoloader
php artisan migrate --force
php artisan config:cache
```

```
php artisan route:cache
php artisan view:cache
```

5. Set up auto-deploy

- Go to **Deployments → Enable Quick Deploy**
- Now, every time you push to your GitHub repo, Forge will auto-deploy

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Laravel Vapor Deployment + CI/CD Pipeline (AWS Serverless)

1. Install Vapor CLI

```
composer global require laravel/vapor-cli
```

Add this to your path:

```
export PATH="$HOME/.composer/vendor/bin:$PATH"
```

2. Login to Vapor

```
vapor login
```

3. Initialize Vapor project

```
vapor init
```

This will generate a vapor.yml file like:

```
id: 1234
name: your-project
environments:
  production:
    memory: 1024
    cli-memory: 512
    runtime: php-8.2
    database: your-db
    cache: your-cache
    domain: your-domain.com
    build:
      - 'composer install --no-dev'
      - 'php artisan event:cache'
```

4. Connect Vapor to AWS

- Vapor automatically provisions AWS Lambda, RDS, SQS, S3, etc.
- Use vapor deploy production to deploy manually.

5. Set up GitHub Actions for CI/CD

Create .github/workflows/deploy.yml:

```yaml
name: Deploy to Laravel Vapor

on:
  push:
    branches:
      - main

jobs:
  vapor-deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v2

      - name: Setup PHP
        uses: shivammathur/setup-php@v2
        with:
          php-version: 8.2

      - name: Install Composer Dependencies
        run: composer install --no-dev --prefer-dist --optimize-autoloader

      - name: Deploy to Vapor
        run: vapor deploy production --commit="${{ github.sha }}"
        env:
          VAPOR_API_TOKEN: ${{ secrets.VAPOR_API_TOKEN }}
```

Set VAPOR_API_TOKEN in your GitHub repo settings → Secrets → Actions

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Summary

Feature	Laravel Forge	Laravel Vapor
Hosting Type	VPS (EC2, DO, Linode, etc.)	Serverless (AWS Lambda, S3, etc.)
Pipeline Integration	Forge UI + GitHub Auto Deploy	GitHub Actions + Vapor CLI
Use Case	Traditional Web Hosting	Scalable, Serverless Architecture
CI/CD Tool	GitHub/Forge built-in	GitHub Actions + vapor-cli

Need a Laravel CI/CD Pipeline on Jenkins, GitLab CI/CD, or AWS CodePipeline?

Let me know your preferred tool or cloud provider. I can give you a full CI/CD pipeline setup for that too.

Would you like a ready-made GitHub Actions workflow or Forge deployment script for your repo?